

Endogenous fertility in pre-transition England

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When did fertility become a choice? An influential literature treats fertility control as a behavioural innovation of the late nineteenth century, enabling greater investment in child quality. However, evidence from a lottery in London that exogenously changed marriage timing suggests that married couples practised fertility control long before fertility fell. Birth intervals lengthened in response to earlier marriage, offsetting the higher fertility that would be expected in the absence of control. This suggests marriage timing and contraceptive effort were substitutes. Family planning was thus present, but demographic change required shifts in incentives to mobilise this latent capacity.

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After centuries of high fertility, the mean number of children born to a woman in England declined from roughly five to two for cohorts born between 1830 and 1900 (Guinnane, 2011). A recent estimate suggests that declining fertility accounts for nearly 70 per cent of the annual growth of GDP per capita in the period 1876-1935 via increased human capital investment, savings, female labour force participation, and changes to the population age structure (Madsen, Islam and Tang, 2020). The fertility transition represents a major inflection in the history of human welfare.

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Although fertility in England had fluctuated with changes in food availability and marriage age (Scott and Duncan, 1999), the fertility transition was marked by a sharp decline in the fertility of married couples (Szreter, 1996; Woods, 2000; Guinnane, 2011). Opinion is divided on whether this represented a behavioural ‘innovation’ (Carlsson, 1966; Alter, 2019). Many regarded birth control as morally offensive or were simply ignorant (Folbre, 2009; Malthus, 1909). For example, Francis Place (b. 1771), a Malthusian who published one of the earliest pamphlets to advocate family planning in England, had fathered 15 children after an early marriage and claimed to have never conceived of birth control until a colleague brought contraceptive sponges to London from France in 1818 (Cook, 2004; Miles, 1988).

It is fitting that Place identified France as the origin of this knowledge transfer, as the French fertility transition inspired an influential theory of fertility decline as a primarily cultural phenomenon (Doepke et al., 2023). Observing that fertility decline began in revolutionary France, not with the industrial revolution in England, and that the timing of fertility decline was only weakly related to economic growth in Europe, the Princeton European Fertility Project concluded that cultural norms and ignorance caused high fertility. The ‘natural fertility’ hypothesis holds that pre-transition fertility lay ‘beyond the calculus of conscious choice’ (Coale, 1973; Coale and Treadway, 1986). This hypothesis is regularly evoked to characterise fertility in the past, particularly by scholars whose work is informed by evolutionary theory (e.g. Colejo-Durán et al., 2024; Dillon et al., 2024; McFadden, 2023; Clark, 2007). A survey of the field of evolutionary demography, for example, characterised the view ‘that human fertility behaviour is driven by conscious decision-making’ as ‘a very big assumption’ (Sear, 2015).

Recent quasi-experimental evidence supports the natural fertility hypothesis directly. Clark, Cummins and Curtis (2020) found that twin births increased average family size by one, suggesting no compensating changes to fertility and leading the study’s authors to conclude that it is possible to regard ‘all the variation in family size as exogenous’ in pre-transition families (Clark, Cummins and Curtis, 2020). However, while monozygotic twins occur more or less randomly, the same biological processes that make dizygotic

twinning more likely also increase overall fecundity (Tong and Short, 1998). It is not possible to distinguish these twin types in historical data. Additionally, twin pregnancies are at greater risk of miscarriage and death, particularly in developing countries where maternal health is likely poor (Guo and Grummer-Strawn, 1993; Marco-Gracia, 2024). Mothers of twins who survive to baptism may also be selected on health.¹ While this study is convincing, without measuring underlying health, fecundity, and zygosity, it is not possible to completely rule out confounding factors.

Further evidence comes from a wide and varied literature documenting the contribution of norms and information to fertility transitions. For instance, secularization and social change roughly coinciding with the French Revolution (c. 1789) were associated with lower fertility in parts of France that had not experienced significant industrialisation (Blanc, 2024; Blanc and Wacziarg, 2020; de la Croix and Perrin, 2018; Perrin, 2022). While evidence of a direct cultural transmission between France and England is speculative (Clark and Cummins, 2015), migrants both carried and transmitted reproductive behaviours during the historical fertility transition (Beach and Hanlon, 2023; Melki et al., 2024). Places more culturally similar to France adopted fertility control sooner, suggesting possible information diffusion along cultural lines (Spolaore and Wacziarg, 2022). For instance, fertility declined faster in French-speaking Wallonia than in Flemish Belgium, and spatial proximity to francophones accelerated Flemish demographic change (Lesthaeghe, 1977; Van Bavel, 2004*b*). Cultural norms and information diffusion also played a role in later fertility transitions in, for example, Germany (Braun, Franke and Öztürk, 2025), North America, South Africa (Beach and Hanlon, 2023), Latin America (Moorthy, Iyer and Moyano, 2025), and Bangladesh (Munshi and Myaux, 2006), and likely contributed to the recent emergence of below-replacement fertility rates in the U.S. (Bailey, 2025; Goldin, 2021). This literature provides an exogenous channel for the emergence of contraceptive behaviours.

On the other hand, many studies have linked longer birth spacing to temporary income shocks, suggesting possible fertility control (van Bavel, 2004*a*; Bengtsson and Dribe,

¹I thank James Fenske for pointing this out.

2006; Bengtsson and Quaranta, 2025; Péter Őri and Levente Pakot, 2025; Marco-Gracia, 2019; Cinnirella, Klemp and Weisdorf, 2017). While such unexpected shocks can help identify causal effects, they do not directly address the question of endogenous lifetime fertility, which remains of central theoretical interest, although some studies have found longer-term effects as well (Cilliers, Mariotti and Martins, 2024; Sandström and Vikström, 2015). Critics argue the methodology for detecting spacing (Cox proportional hazards) in historical populations may be particularly prone to specification errors and truncation bias (Clark and Cummins, 2019; Cinnirella, Klemp and Weisdorf, 2019; Alter, 2019). Furthermore, because income may also affect birth intervals through health, fecundity, and breastfeeding duration (Oris, Mazzoni and Ramiro-Fariñas, 2024; McFadden, 2023), which are rarely observable in historical datasets, these findings are also vulnerable to omitted variable bias (e.g. Klesment and Lust, 2025; Willführ and Perez, 2025).

This study also considers spacing as a birth-control technique but makes two novel methodological contributions. First, I study an historical lottery to bypass the problem of biological unobservables via treatment randomization. Raine’s charity operated a semi-annual lottery for a small group of unmarried women in London between 1758 and 1872; the winner received £100 (equivalent to roughly £19,040 today) on condition that she marry within six months.² Second, although other studies demonstrate birth control via birth spacing, I demonstrate that birth spacing also responded to marriage timing—the primary mechanism affecting lifetime fertility before the fertility transition (Woods, 2000). Differences of opinion also exist about whether marriage timing responded endogenously to planned fertility or was exogenously affected by labour market conditions (Sarti, 2008; Horrell, Humphries and Weisdorf, 2020; Szreter and Garrett, 2000; Foreman-Peck, 2011; Voigtländer and Voth, 2013). I find that lottery winners married earlier, but this had no impact on their fertility due to compensating change to birth spacing. Because lottery winning is plausibly uncorrelated with unobservable biological determinants of birth spacing, this suggests deliberate control. More broadly,

²Conversion uses reconstructed purchasing power, see (Measuringworth.com, 2024).

the findings imply that marriage timing and birth spacing may have been substitutes for achieving fertility outcomes, thus linking existing research on family planning via birth intervals to more general modes of fertility behaviour (e.g. marriage timing).

This study contributes to three literatures. First, unified growth models that assume endogenous fertility before the demographic transition can cite these findings in support of this assumption (e.g. Galor and Weil, 2000; Le Fur and Wasmer, 2025; Cervellati, Meyerheim and Sunde, 2023; Foreman-Peck, 2011). However, the presence of direct fertility control suggests that the fertility–income relationship may have been less constrained in pre-transition populations than in some versions of the theory, such as Galor and Moav (2002), which model a positive fertility–income gradient arising from the relaxation of a subsistence constraint.³ Second, the family is an important economic institution, and this study may contribute to a reading of the historical family as rational and the fertility transition as an example of induced institutional change (North, 1990; Gay, Gobbi and Goñi, 2026; Anderson and Bidner, 2023; Lundberg and Pollak, 2007; Pollak, 1985). Third, the findings add to the literature on family planning policy in developing economies by showing that birth control was feasible even in the information– and contraceptive technology–poor setting of eighteenth-century London (de Silva and Tenreyro, 2020; Bongaarts, 2020; Cavalcanti, Kocharkov and Santos, 2021; Miller and Babiarz, 2016).

Section one describes the lottery and its historical context. Section two develops a parsimonious model to interpret the effect of the lottery on fertility. Section three describes the dataset, its representativeness, and imputations used in its construction. Sections four and five describe the identification strategy and present results. Section six presents complementary evidence from feast days. Section seven concludes. This study provides novel empirical evidence of endogenous pre-transition fertility in the first population to experience an industrial revolution and sustained economic growth.

³This complements Davenport’s (2019) research on infant mortality in the same historical setting, which finds a positive or flat wealth–infant mortality gradient.

I. Historical Setting

Henry Raine (1679-1738) was a brewer who made a considerable fortune quenching the thirst of sailors in East London's dockland (Lincoln, 2018; Cockburn, King and McDonnell, 1969). Simultaneously, he was an active and devout member of the Church of England. Raine apparently resolved the contradiction between his pious spirit and his profane livelihood through charitable acts, including a school established in 1719 that admitted boys and girls. Raine's was part of an evangelical wave of charity school foundations sweeping over London in the early eighteenth century in reaction to perceived irreligion among the poor (Rose, 1991; Jones, 1964). In addition to learning to read bible verses, however, girls who attended Raine's school had a chance to win the 'marriage portion'.

Although his precise motives are unclear, it seems that Raine introduced the marriage portion out of a similar concern for maintaining church membership, as his will required that both bride and groom were members of the Church of England (Rose, 1991). Per Raine's instructions, girls had to go through a number of steps before they could be eligible for the prize (Raine, 1748). First, to be admitted to the school, six local residents needed to vouch for their character and respectability. At the same time, the school's trustees had a mandate to admit children of poor families in the parish of St. George in the East who could not otherwise afford school fees. These combined constraints meant school children likely came from the households of local artisans or sailors who adopted middle-class respectability without the standard of living to match (Rose, 1991). For instance, Ann Cater's admission record in 1822 noted simply, 'Mother dead, father left with 7 small children'.⁴ Next, girls were selected from the lower school to enter the upper 'asylum', where they were taught skills relevant to eventual employment in domestic service, which the school arranged (Cockburn, King and McDonnell, 1969). Finally, subject to a positive character reference from their employers, these women could step forward to claim the marriage portion after their twenty-second birthdays.

⁴TLA ACC/1811/8/11/1.

These conditions were not extraordinary for the time period despite their apparent strictness. For example, the Church of England, the state church, accounted for 49 per cent of all church attendances in 1851 and was the single-largest denomination, making a large pool of potential lottery participants (Snell and Ell, 2004). At a time of high dependency ratios, the average family experienced life-cycle poverty when young children were present in the household, making also many potential candidates for charity (Horrell, Humphries and Weisdorf, 2022). Further, domestic service was a common experience for young girls. It was the largest occupational group in the nineteenth and possibly eighteenth centuries, employing as many as 40 per cent of all women in 1851 (You, 2024). There is no sign that these girls were employed in exceptionally ‘elite’ households. The school briefly recorded girls’ wages in service between 1780 and 1790. Their median yearly earnings were £3, far below the £7.35 median wage earned by other domestic servants in London at the same time.⁵ Further, character references were common in the labour market for domestic servants (Kaiser, 2025). Thus while it was unusual for girls to have their lives so thoroughly shaped by a charitable institution, the shape those lives took was not.

Table 1 compares the occupations of fathers of girls who participated in the lottery to a random sample of fathers’ occupations taken from the baptismal registers of St. George in the East between the years 1730-1840, roughly corresponding to the birth cohorts that could have been eligible for the portion. The table reports odds ratios and standard errors estimated from a series of logistic regressions of occupational class (HISCLASS) on a dummy variable indicating lottery participation. In general, lottery participants were more likely to come from semi-skilled and artisan families and slightly less likely to come from unskilled families. However, they were also much less likely to come from professional families and those of retailers and wholesalers trading on their own account. This agrees with the intuition above that lottery participants came from modest backgrounds and were objects of charity more due to bad luck than destitution.

Twice yearly, up to six women could stand for the marriage portion. They drew sealed

⁵TLA ACC/1811/8/14; London average from data in Kaiser (2025), kindly shared by the author.

TABLE 1—SAMPLE REPRESENTATIVENESS

HISCLASS	Odds Ratio	S.E.
12 – Unskilled farm workers	1.05	(1.29)
11 – Unskilled workers	0.81	(0.19)
10 – Lower-skilled farm workers	1.05	(0.92)
9 – Lower-skilled workers	1.88	(0.63)
8 – Farmers and fishermen	-	-
7 – Medium-skilled workers	1.56	(0.37)
6 – Foremen	-	-
5 – Lower clerical and sales personnel (low skill)	0.29	(0.32)
4 – Lower clerical and sales personnel (medium skill)	0.36	(0.18)
3 – Lower managers	0.29	(0.32)
2 – Higher professionals	2.11	(2.99)
1 – Higher managers	-	-

Note: Estimated from a series of logistic regressions of occupational class on a dummy variable indicating lottery participation. The comparison group was a random sample of fathers' occupations from the St. George in the East parish registers for 1730-1840. The sampling scheme took the first occupation on every fifth page from 1730-1812 and every twentieth page after 1812, when register entries became lengthier. This scheme resulted in approximately three random occupations per year. These were classed using the HISCLASS schema. Missing values indicate the absence of that class in at least one of the comparison groups.

Source: The London Archive (2010a)

tickets from a tin canister, one of which was marked. The candidates simultaneously opened their tickets, revealing the winner to the assembled public. The winner was then allowed six months to find a suitable groom-to-be, whose character was also evaluated by the trustees. Eligible grooms needed to be resident in St. George in the East or two neighbouring parishes. The couple was then paid £100 on their wedding day. According to one trustee, most women had suitors at the time of the draw, but this was not always the case (Jones, 1875). If she was unable to find a groom after six months, the winning candidate received only £5 and became ineligible for future draws. Women who stood for the prize but drew blank tickets were allowed to re-enter subsequent draws 'so that every every one of them may happen, at one Time or other, to be elected, and entitled to such Sum of One Hundred Pounds for a Marriage Portion' (Raine, 1748). After the draw, £5 were expended on a wedding feast.

These terms were set out in Raine's will in 1736. However, because Raine had only endowed the fund with £4,000 in 3 per cent gilts, it was left to accumulate until it yielded the required £210 per year. This apparently occurred in 1758, when the first marriage lot-

tery is recorded in surviving archival registers (Cockburn, King and McDonnell, 1969). This delay created a large pool of eligible women at the beginning of the lottery, making the early draws more competitive. Indeed, between 1758 and 1782, more than the maximum allowable six women often sought to participate in the lottery. Trustees carefully made note of the candidates, who became eligible for subsequent lotteries in the order they had signed up. After the initial glut cleared, lottery participation varied, apparently in line with the fortunes of the school, with a normal lottery consisting of around three participants (figure A1).

The last competitive lottery occurred in 1872, after which time the number of applicants declined, often leaving only one candidate for each lottery. This decline may have related to wider policy changes occurring at the time. The Endowed Schools Act of 1869 created a commission with wide powers to intervene in the administration of secondary schools, and trustees felt their rights threatened. To try to head off forced reform, the boys school voluntarily dropped elementary teaching and developed its secondary-level curriculum for fee-paying students, with the charity providing scholarships to examination candidates. One trustee of the charity expressed the urgency of reforming the marriage portion as well, and the girls' asylum appears to have admitted far fewer girls after 1869 (Jones, 1875). The asylum ultimately closed in 1883 (Cockburn, King and McDonnell, 1969). During its functional lifetime, then, the charity executed approximately 228 marriage lotteries according to the system described in Raine's will.

St. George in the East was a large docklands parish closely linked to the old Port of London. It had grown up in the seventeenth century, and although port expansion continued rapidly to the east, its population grew at a relatively modest pace over the eighteenth century (Marriott, 2011). There appears to have been nothing extraordinary about the parish's fertility rate. Figure 1 graphs the crude fertility rate (CFR) from 1850, when local civil registration data become available, to 1910. The CFR in St. George in the East is slightly higher than the average for England and Wales. However, because mariners would possibly have been away at sea on census night, this may be a partial artifact of underestimation in the denominator. Equally, there is no indication that the

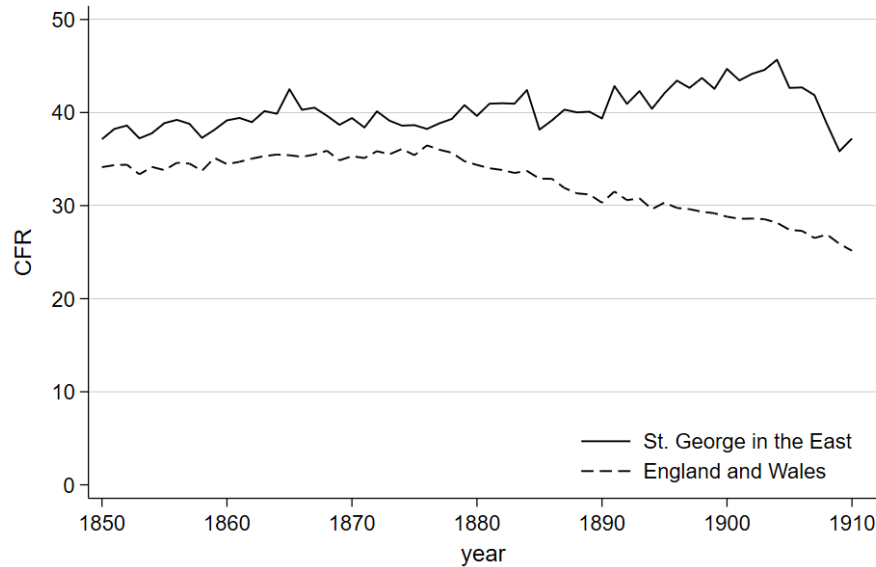


FIGURE 1. CRUDE FERTILITY RATE IN ST. GEORGE IN THE EAST, 1850-1910

Note: The crude fertility rate (CFR) is the number of births per 1,000 population. Births are reported yearly by the registrar general, while population is linearly interpolated between decadal censuses.

parish was an early participant in the fertility transition. While the national CFR begins to decline in the 1870s, there is no sign of decline in St. George until possibly 1905.

Finally, I note that innovation in contraceptive technology does not explain the fertility transition. While physical contraceptives did exist, they were not marketed or consumed in significant numbers until the early twentieth century (Jones, 2020; Youssef, 1993). Changes to the frequency of sex and its distribution within marriage are sufficient to account for variation in fertility in this period (Szreter, 1996). For example, Stanford and Dunson (2007) show that a reduction in the frequency of intercourse from twice per week to once per week can increase the expected duration of the birth interval by 61 per cent if intercourse occurs randomly throughout the menstrual cycle.⁶ Some couples practiced *coitus interruptus*, but women might also adopt strategies to reduce the frequency of intercourse without requiring male buy-in, such as ‘staying up late at night working,

⁶Own calculations from authors’ model.

sharing beds with children, complaining of pains, or ... enlisting the doctor's support', as Cook (2004) documents. Such methods were equally viable before and after the fertility transition.

Taken together, these features underscore that the institutional setting was distinctive, but the demographic context was not. What matters for the analysis is how these ordinary women responded to extraordinary incentives and what this reveals about their reproductive agency. The next section develops a framework to evaluate Raine's lottery in this light.

II. Theory

In this pre-transition setting, one unresolved issue is the relationship between marriage timing and fertility (Szreter and Garrett, 2000). Theoretical approaches to this question emphasise how marriage timing relates to the 'gains to trade' when spouses divide household tasks (Keeley, 1977). These are themselves affected by labour market structure, particularly married women's access to careers, the marginal utility from children and child quality, contraceptive technology, and institutional factors affecting, for example, divorce (Greenwood, Guner and Vandenbroucke, 2017; Doepke et al., 2023). However, in Hruschka and Burger's (2016) study of 200 high-fertility populations, the key stylised fact to emerge is that pre-transition fertility closely resembles a Poisson process, implying a relatively constant risk of childbirth over the reproductive span. If childbirth occurs primarily in wedlock, this implies relatively constant fertility over marriage. This may be due to the nature of contraception, discussed above, or because factors that motivate clustering births were largely absent. For example, in nineteenth-century England, returns to experience for women were negligible in both textile factories and agriculture (Burnette, 2006; Boot, 1995). On the other hand, a constant risk of childbirth would also be expected under the natural fertility hypothesis. The remainder of this section develops a parsimonious model to help distinguish natural from endogenous fertility empirically in the context of Raine's lottery.

Consider a woman choosing when to marry. She chooses $t \in \{0, 1, 2, \dots, T\}$, where t

is the number of unmarried periods and T is her adult lifetime. This implies $T - t$ is the duration of marriage. For simplicity, assume no extramarital births so that her duration of marriage determines her time available for reproduction.

Take first the natural fertility case where women do not control fertility within marriage. Although in reality, fertility declines naturally with age, this is left out of the model for simplicity (Henry, 1961). Births therefore arrive stochastically within marriage at a constant rate, λ . The expected number of births in a marriage of duration $T - t$ is therefore

$$(1) \quad E(N) = (T - t)\lambda.$$

Delaying marriage by one period will reduce expected family size by λ .

Under endogenous fertility, the woman may exert contraceptive effort e to reduce the probability of a birth, $\partial\lambda/\partial e < 0$. Although she may choose the level, I assume that she does not vary her contraceptive effort over time within marriage. This greatly simplifies the problem and reflects the stylized facts discussed above.⁷ Assuming no discounting, lifetime expected utility will therefore be

$$(2) \quad E(U) = ts - (T - t)c(e) + E(v[N, \lambda(e)])$$

where s is the per-period utility flow from being single and $c(e)$ is the utility cost of contraceptive effort in marriage, increasing in contraceptive effort ($\partial c/\partial e > 0$). Lifetime utility increases in completed family size ($\partial v/\partial N > 0$; $\partial v/\partial^2 N \leq 0$), and wider birth spacing (lower λ , i.e. $\partial v/\partial \lambda < 0$) to reflect a quantity-quality trade-off.⁸ This feature of the model is intended to capture the association between longer birth spacing and better infant health via, for example, reduced breastfeeding-pregnancy overlap (Dadi, 2015;

⁷Dynamic fertility models in which contraceptive effort is allowed to vary across multiple periods often have no closed-form solution without making strong assumptions about the functional form of the utility function. See Arroyo and Zhang (1997).

⁸The separation of utility flows from lifetime utility derived from final family size is similar to Dioikitopoulos and Varvarigos (2023).

Conde-Aguedelo et al., 2012).⁹ Although t and N are discrete, I treat them as continuous and differentiable for tractability.

The woman chooses when to marry and her level of contraceptive effort, which determines her expected family size. To simplify the expectation operator in (2), I approximate the function with a second-order Taylor series evaluated at $E(N)$

$$\begin{aligned} E(U) &= ts - (T - t)c(e) \\ &\quad + E \left[v[E(N), \lambda(e)] + v_1[N - E(N)] + \frac{1}{2}v_{11}[N - E(N)]^2 \right] \\ E(U) &= ts - (T - t)c(e) \\ &\quad + v[E(N), \lambda(e)] + \frac{1}{2}v_{11}\text{Var}(N), \end{aligned}$$

where subscripts denote partial derivatives with respect to the relevant argument. It is then possible to rewrite the maximization problem in terms of (1)

$$(3) \quad \max_{t,e} U(t, e) = ts - (T - t)c(e) + v[(T - t)\lambda(e), \lambda(e)] + \frac{1}{2}v_{11}(T - t)\lambda(e).$$

The first-order conditions are:

$$\begin{aligned} (4) \quad & \lambda(v_1 + \frac{1}{2}v_{11}) = s + c(e) \\ (5) \quad & v_2\lambda_1 = (T - t)(c_1 - \lambda_1[v_1 + \frac{1}{2}v_{11}]). \end{aligned}$$

The first, (4), says that the instantaneous utility of another period of marriage in terms of children must be equal to that of another period of singledom and the disutility of contraceptive effort. The second, (5), says that the advantage of contraceptive effort on spacing must offset the disadvantages of fewer children and the disutility of contraception in marriage. The second derivative reflects concavity in the utility derived from realised fertility and can be conceived as risk aversion over the expected number of children. I

⁹Short birth intervals also impact infant health via maternal health. This parameter may therefore reflect preferences over maternal health. For modelling purposes, this parameter serves primarily to ensure that the model does not always collapse into a corner solution with no spacing, and either assumption serves this purpose.

assume that risk-aversion is modest so that

$$(6) \quad v_1 + \frac{1}{2}v_{11} > 0,$$

which prevents a contradiction in the first-order condition on marriage timing (4). Economically, this would correspond to a corner solution in which risk aversion in the number of children is so great that it is better to avoid marriage altogether.

The two FOCs characterise the optimum when the woman chooses both t and e . To analyze Raine's lottery, I now consider the case where t is fixed exogenously and effort adjusts endogenously. From (1), expected fertility in this scenario is

$$(7) \quad \begin{aligned} E(N | t) &= (T - t)\lambda[e(t)] \\ \frac{\partial E(N | t)}{\partial t} &= -\lambda + (T - t)\lambda_1 \frac{\partial e}{\partial t} \end{aligned}$$

The first term is the mechanical effect of losing a period; the second term captures behavioral adjustment. Relative to the exogenous case, the impact of a shock to marriage timing will depend on the sign of $\partial e / \partial t$. If later marriage reduces contraceptive effort ($\partial e / \partial t < 0$), i.e., if marriage timing and contraceptive effort are substitute strategies, then the fertility penalty of delay is smaller than under natural fertility.

For marriage timing and contraceptive effort to be substitutes requires assumptions about the concavity of utility with respect to realised fertility. The second FOC, (5), describes how optimal effort responds to a change in timing

$$F(e^*(t), t) = v_2\lambda_1 - (T - t)(c_1 - \lambda_1[v_1 + \frac{1}{2}v_{11}]) = 0.$$

Note that because the lottery exogenously shifts t , the expected number of births, $E(N) = (T - t)\lambda$, varies across the comparative static, and the derivatives of v evaluated at the new $E(N)$ must be treated as functions of t .

Implicit differentiation gives the relationship between optimal effort and marriage tim-

ing

$$(8) \quad \frac{\partial e^*}{\partial t} = -\frac{F_2}{F_1},$$

which will be negative (i.e., consistent with the idea that effort and timing are substitutes) if F_2 and F_1 have the same sign. By the second-order condition for a maximum, $F_1 = \frac{\partial^2 U}{\partial e^2} < 0$. The sign of $\partial e^*/\partial t$ thus depends on the sign of F_2 ,

$$F_2 = -v_{21}\lambda_1\lambda + (c_1 - \lambda_1[v_1 + \frac{1}{2}v_{11}]) - (T-t)\lambda_1\lambda(v_{11} + \frac{1}{2}v_{111}).$$

I simplify by assuming that utility is separable in realised family size and birth spacing. This eliminates the cross-partial. Further, in line with the second-order Taylor approximation, I drop the third-order derivative. This is equivalent to assuming v is well-approximated by a quadratic in N over the range of $E(N)$ induced by variation in t . The simplified expression is therefore

$$(9) \quad F_2 = (c_1 - \lambda_1[v_1 + \frac{1}{2}v_{11}]) - (T-t)\lambda_1\lambda v_{11}.$$

Marriage timing and contraceptive effort will be substitutes if this expression, which is the cross-partial of utility with respect to effort and timing, is negative. Replacing variables assumed to be negative with their absolute values times -1 and rearranging, this implies

$$c_1 + |\lambda_1|(v_1 - \frac{1}{2}|v_{11}|) < (T-t)|\lambda_1||v_{11}|\lambda.$$

Rearranged and substituting (1), this becomes

$$(10) \quad |v_{11}|\left(E(N) + \frac{1}{2}\right) > \frac{c_1}{|\lambda_1|} + v_1.$$

To build intuition, recall that F_2 measures how the marginal utility of effort varies with marriage timing. The costs of effort are twofold. There is a direct cost via c_1 and an indirect cost via v_1 , reduced fertility. With shorter marriage, the direct cost is incurred over

fewer periods, which reduces the marginal cost of effort. More subtly, when marriages are shorter each unit of effort is less costly in terms of its marginal impact on expected fertility. These effects are reflected on the right-hand side of the inequality. The left-hand side reflects the countervailing effect of change in the marginal utility of fertility in levels. With later marriage, expected fertility falls, and this increases the marginal utility of an additional child. When this effect predominates, effort and timing will be substitutes in the sense that effort is reduced (increased) to counteract the decline (increase) in expected fertility due to later (earlier) marriage.

The curvature of utility over realised fertility can be further characterised using (6), which provides an upper bound on concavity for an interior solution

$$(11) \quad |v_{11}| < 2v_1.$$

Combined with the lower bound in (10), it follows that concavity must fall in the interval

$$(12) \quad \frac{v_1 + \frac{c_1}{|\lambda_1|}}{E(N) + \frac{1}{2}} < |v_{11}| < 2v_1$$

if substitution is to hold at an interior solution.

To get substitution between effort and timing, this implies that concavity in the utility derived from fertility should not be too high or too low. Specifically, the set of values of $|v_{11}|$ will be non-empty if $v_1 + \frac{c_1}{|\lambda_1|} < 2v_1(E(N) + \frac{1}{2})$. This is an open interval over a plausible range of values. Rearranging the inequality gives

$$(13) \quad E(N) > \frac{c_1}{2v_1|\lambda_1|}.$$

This says that expected fertility must exceed a threshold set by the ratio of the direct cost of effort to the marginal utility of a child weighted by the effectiveness of contraception.

This condition will tend to hold when expected fertility is high, the cost of effort is low, children are valuable, and contraception is effective. This is plausible for high-fertility populations, but it implies that substitutability between marriage timing and effort may

fail if expected fertility declines or if the utility of child quantity declines, possibly because a quantity-quality trade-off shifts in favour of quality. Both conditions have been theoretically linked to demographic transition, which also coincided with changes in fertility behaviours. The concavity of utility in expected fertility provides a theoretical connection. As expected family sizes fell, marriage timing and contraceptive effort shifted from substitutes to complements. Indeed, during the early stages of the fertility transition in England, both later marriage and a decrease in marital fertility contributed to fertility decline (Woods, 2000).

Finally, I consider the corner solution in which the woman exerts as little contraceptive effort as possible. Qualitatively, this will be very similar to the natural fertility hypothesis. In the corner solution, effort cannot adjust to marriage timing because it has already reached its floor, whereas under natural fertility, effort cannot adjust because the behavioural mechanism does not exist. The corner solution will emerge in the model when the marginal utility of birth spacing is low. As discussed, contraceptive effort has a direct cost reflecting the effort or foregone pleasure associated with family planning and an indirect cost via lower fertility. The only good provided by contraceptive effort in the model is birth spacing. If this good is only weakly valued, the no-effort corner solution will tend to dominate.

The model therefore carries three important implications. First, if the relationship between fertility and marriage timing is weaker than expected under natural fertility, marriage timing and contraceptive effort must be substitutes. As corollaries, this implies that the cost of effort is low relative to the utility of children and that households derive utility from spacing.

III. Dataset construction and validation

To study fertility outcomes among lottery participants, I hand-link information from manuscripts created during the administration of Raine’s charity to data on life events contained in online genealogical databases (e.g. Ancestry.com and Findmypast.co.uk). Relative to automated linking, following a recent critical survey (Bailey et al., 2020),

hand linking would be expected to produce links of the highest possible quality and minimize bias, and it is viable given the relatively small size of the dataset.

Two primary sources of information on the lottery are held in The London Archives: lot books and trustees' receipts (The London Archive, 1736). The lot books were created during the marriage portion ceremony, while the receipts record the marriage-portion transaction and provide a useful cross-reference for the lot books.¹⁰ From these records, I note for each lottery the date of the draw, a list of candidates, the number of times each candidate participated, the winner, the name of her groom, the groom's occupation, and the date of their marriage. I then link each candidate to her school admission register, which notes her date of birth and often her father's name and occupation.

For lottery participants, I thus possess relatively rich pre-treatment information, but the quality of post-treatment information varies. The manuscripts provide no information about the marriages of women who either drew the prize but did not marry within six months or who dropped out of the lottery without winning. This introduces a correlation between treatment and data quality whose implications for causal identification are left to the next section. Nonetheless, I search for these women in the collection of all London parish marriage registers digitized by The London Archive to obtain their spouse's name and their date of marriage (The London Archive, 2010*a,b*). I restrict search to the three years immediately following the candidate's departure from the lottery and only accept links if the bride's name is unique within that window.

Next, I look for evidence of childbearing in the London baptismal records, which are also digitized by The London Archives. I restrict my search to the 30-year period after marriage. Here, I link on spouse-parent names and rely on rule-of-thumb tie breaking. Where two sets of parents share identical names, I favour those whose children were born shortly after the wedding date. Further, because mother and father must both have lived in or near St. George's parish to be eligible for the lottery, I favour matches living in East London. Finally, I use the father's occupation if this seemed to provide identifying information. That is, I interpret skilled occupations that likely required an apprentice-

¹⁰The two-hundred-plus-year-old documents were occasionally illegible due to wear and tear.

ship as providing reliable information about identity because these are more likely time invariant. Where two potential matches have occupations that are closely related or in the same industry, I do not rely on this information to break ties. Where none of these rules of thumb provide grounds for disambiguating a match, I make no match and drop the couple from the sample.

Once I have identified the first child's baptismal record and birth date, another baptism usually follows within roughly two years. Where there is a sequence of baptisms of this kind, and none of the family's other details change, I am confident that I am identifying siblings. If there is a break in the sequence which starts up again roughly two years later in a nearby parish and the father's occupation has not changed, I also record these new children as siblings. When possible, I cross-reference against the decennial census, which began in 1841. In some cases, a child is only identified in the census and not in the baptismal records. I also include such children and subtract their reported age in years from the census date to reconstruct their approximate birth date. It was not always possible to unambiguously distinguish the children of one family from another, and such cases were dropped from the sample.

The resulting dataset is imperfect, but it represents an earnest effort at accuracy given the recognised challenges of linking across historical sources in London (Davenport, 2016).¹¹ The manuscript sources provide a unique means of verifying the accuracy of my reconstitution method. In 1851, the trustees wrote to all marriage portion recipients of the last decade and preserved some of their correspondence. If the trustees were able to locate the couple, they noted a residential address and the occupation of the groom. I have reproduced all identifying information from these manuscript sources in Table 2 compared to the information I obtained from the 1851 census and the baptismal record of the child born nearest to 1851. This is a blind validation exercise, as I did not draw on this source in constructing the dataset.

In table 2, type I errors refer to cases where I have made a link that does not match

¹¹An annotated dataset, including direct URLs to the sources, is available on request to be checked by interested readers.

TABLE 2—VALIDATION OF RECONSTITUTION METHOD, 1841-1851

Manuscript		Census		Baptism		Type I	Type II
5 New Street, Horsley-down	Fellmonger	—	—	New Street, Horsleydown	Fellmonger		
15 Tottenham Place, Tottenham Court Road	Baker	15 Tottenham Place	Baker	Upper North Place, St Pancras	Baker		
Red Lion Passage	Pastry cook and confectioner	—	—	—	—		X
Unknown	—	—	—	Old Montague Street, Whitechapel	Bricklayer ¹	X	
2 Morpeth Street, Bethnal Green	Bell founder	Morpeth Street, Bethnal Green	Bell founder	Bethnal Green	Bell founder		
Unknown	Optical brass founder ²	—	Brass finisher	—	Brass turner		
St. Katharine Docks	Fireman	—	—	—	—		X
10 Norfolk Street, Commercial Road	Gun Maker	New Norfolk Street, Stepney	Gun polisher	7 [illegible] Cornwall St	Gun maker		
Unknown	Shoemaker	—	—	—	—		
4 Little Abbey Street, Bermondsey	Silk weaver	4 Stephen Street, Bermondsey	Weaver	—	—		
Unknown	Shoemaker	27 St. James Terrace	Shoemaker journeyman	11 Tarling Street, Christ Church	Bootmaker		
3 Hope Place, Bermondsey	Warehouse man	3 Hope Place, Bermondsey	Porter	New Church Street, Bermondsey	Porter		
5 Curriers Hall Court, London Wall	Porter	9 Three Herring Ct, Cripple-gate	Porter	Marshall St., Gripple-gate	Porter		
Unknown	Shipmate	12 Prospect Place	Mariner	12 Prospect Place, St George in the East	Mariner		
Went abroad	Painter	—	—	—	—		
Unknown	Shoemaker	19 Lombard St, Chelsea	Shoemaker	16 Lombard, Chelsea	Cordwainer		
Unknown	Cooper	Denmark Street, St George in the East	Cooper	7 Denmark Street, St George in the East	Cooper		
19 Catherine St, St. George East	Oil & Colour-man	—	—	27 Fenton Street, St George in the East	Colourman		

Note: 1: Groom was described as bricklayer at time of his marriage. 2: Mother of bride wrote to school to say her daughter had died, and husband was reported as a widower in 1851 census.

Source: See text.

the trustees' correspondence. However, because high-frequency, short-range mobility was common in London at this time (Davenport, 2016), I allow for some geographical mobility and do not flag as an error a change of address to another house in roughly the same neighborhood. In only one case, roughly 5 per cent of the sample, have I attributed lottery-winning to a family living in East London that the trustees did not themselves identify. In this case, however, the father had the same occupation, bricklayer, as the groom on his wedding night. It is possible this is no error and the trustees simply lost touch with this family. Type II errors refer to cases where the trustees have located the family, but I have been unable to do so. There are two such cases, representing 11 per cent of the sample. In other cases where I have been unable to make a link the trustees have also been unable to locate the family, indicating possible emigration from London, death and remarriage, or some other complication. I regard such cases as true negatives, not errors. For comparison, hand-linked US census samples have a type-I error rate of at least 4 per cent, while common automated linking methods have a type-I error rate ranging between 15 and 37 per cent and a type-II error rate between 63 and 79 per cent (Bailey et al., 2020).

However, comparing the dataset against other benchmarks indicates it is likely the reconstitution missed some births. The challenge of reconstituting families in London is well-known, stemming from a high prevalence of short-distance migration and the large number of urban parishes, each of which kept vital records of varying quality (Davenport, 2016). If a child died before their baptism, their birth would also tend to go unrecorded, and London's infant mortality rate was high (Wrigley et al., 1997). Table 3 presents summary statistics from the Raine's charity dataset and two comparable historical datasets from England. The first is Davenport's (2016) reconstitution for St. Martin in the Fields, another large urban parish west of the City of London. The second is Wrigley et al.'s (1997) reconstitution of 26 non-urban parishes and towns. The data quality is likely higher in the rural sample because the underlying population is less mobile, but it would miss urban-specific demographic traits if such exist.

The observed number of births in the Raine's lottery sample is much lower than either

TABLE 3—SUMMARY STATISTICS AND COMPARISONS

	mean	sd	min	max
Start Age	23.7	2.15	20.1	29.6
Marriage Age	25.8	2.63	20.8	31.7
Total Bapt.	3.38	2.03	1	9
Final Birth Age	34.7	5.64	23.9	47.3
First Child	29.0	29.7	-4.64	171.2
Middle Child	29.8	16.1	6.51	128.3
Last Child	39.7	21.6	0	114.4
<i>St. Martin in the Fields, 1752-1812</i>				
Total Bapt.	4.43	2.43	2	15
Middle Birth Interval	25.52	12.29	7	127
Last Birth Interval	29.58	14.34	9	119
<i>National sample, 1750-99</i>				
Female Marriage age	24.0			
Completed family size	5.61			
Age at Final Birth	39.3			
First Birth Interval	15.0			
Middle Birth Interval	29.4			
Last Birth Interval	41.0			
<i>After imputation</i>				
Total Bapt.	3.61	1.90	1	9
Final Birth Age	34.4	5.22	23.9	47.3
First Birth Interval	22.7	17.1	1.97	85.6
Middle Birth Interval	29.8	12.1	10.6	70.0
Last Birth Interval	37.6	15.3	0	85.6

Note: Birth intervals in the national sample are only reported for the whole period 1580-1837.

Source: Wrigley and Schofield (1983); Wrigley et al. (1997). Davenport kindly shared data underlying her (2016) article.

the rural or the urban sample. However, because Davenport's methodology depends on observing two subsequent births, singleton households are excluded (Davenport, 2016). The comparable restricted mean ($n > 1$) in Raine's charity is 4.23. This is closer to Davenport's estimate, but still far from the national rural sample. Fertility may have simply been lower in urban settings, possibly due to elevated disease prevalence (Szreter and Siena, 2021).

The pattern of birth intervals in the Raine's dataset is more problematic. In historical fertility data, it is common to find relatively short first birth intervals, longer middle inter-

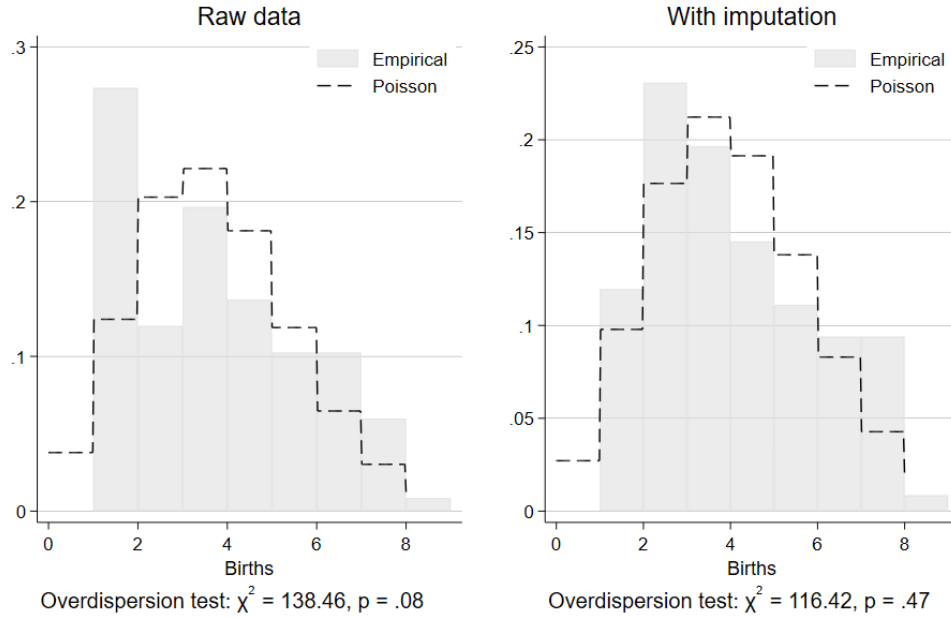


FIGURE 2. EMPIRICAL DISTRIBUTION OF NUMBER OF BIRTHS VERSUS POISSON DISTRIBUTION

vals due to breastfeeding, and longest final intervals due to declining fecundity (Wrigley et al., 1997), but this pattern is absent here. This difference is apparently driven by extreme outliers. Further, the relatively low mean age at final birth in the Raine's dataset suggests some birth histories may be prematurely truncated. Finally, as discussed above, the number of births should roughly follow a Poisson process, but an overdispersion test rejects this hypothesis (figure 2; $p = 0.08$). I provide evidence that these anomalies are more plausibly attributable to missing births than to atypical demographic behaviour, as they vanish after imputing births in long birth intervals.

To identify long birth intervals, I assume births follow a Poisson process and model inter-birth intervals using the exponential distribution, allowing a nine-month offset following multiparous births to account for gestation. I condition the exponential distribution on age at marriage and marriage duration, using means reported by Wrigley et al. (1997) for their rural sample (Table A2). If the cumulative probability of an observed

interval exceeds 0.95 under this distribution (roughly seven years), I treat the interval as inconsistent with the assumed birth process and impute a missing birth at the midpoint. This threshold reflects a probabilistic criterion for identifying gaps in the birth record, rather than a formal hypothesis test.

Despite the inherent limitations of historical record linkage, imputation considerably improves the fit between the data and demographic priors. For example, the lower part of table 3 re-calculates the birth intervals using the imputed data, which now follow the expected increasing pattern by birth order. Further, an overdispersion test now fails to reject the hypothesis that the empirical distribution of family size is Poisson-distributed (figure 2; $p = 0.47$). However, this imputation strategy cannot address truncated birth histories so the age at final birth remains the same. I address this below by defining my dependent variable to minimize the potential impact of truncation. The main analysis is performed using the dataset with imputed births; the appendix reports results for the observed-only births. Results are qualitatively unchanged, but coefficients show signs of minor attenuation bias consistent with greater measurement error without imputation.¹²

IV. Identification

Identification first exploits random assignment in the lottery. Each draw was fair and public, so conditional on entering, the probability of winning was independent of potential outcomes. A potential concern is that participants could drop out after losing and so had partial control over their treatment. To address this, I restrict attention to the first round of the lottery. At this point, no prior draw had occurred, so the outcome is orthogonal to unobserved traits that might otherwise influence both continued participation and fertility. Winning in the first round provides an instrument for earlier marriage because it shifts the timing of marriage without directly affecting fertility, except through marriage timing. This satisfies the exclusion restriction under the assumption that the lottery outcome does not influence fertility through any other channel.

Second, I control for a participant's age at first entry into the lottery. This variable cap-

¹²As a coefficient closer to zero leads to rejection of the natural fertility hypothesis, the imputed births make the main analysis the more conservative of the two.

tures pre-treatment preferences over marriage timing before the draw outcome is known in a manner roughly analogous to a lagged dependent variable in a panel setting. For example, as discussed above, many participants already had suitors when they signed up for the lottery (Jones, 1875). Including initial participation age therefore provides a powerful control for pre-treatment characteristics and preferences that could otherwise confound the relationship between marriage age and fertility.

This conditioning strategy mitigates a shortcoming of the first strategy. As discussed, the manuscript records do not contain information on the spouses of lottery participants who did not win the prize. Treatment thus affects data-quality, which in turn affects the probability of making a successful record link because less information is known about these individuals. Although the selection mechanism therefore operates mainly through the amount of information coded in a name (e.g. name uniqueness), which is plausibly orthogonal to marriage and fertility, this will nonetheless introduce bias if selection also depends on unobservables related to marriage age and fertility (Hughes et al., 2019). Insofar as initial lottery age captures many traits and preferences affecting marriage age, this bias should be minimized.

Nevertheless, selection remains a possible concern. To address this directly, I implement inverse probability weighting (IPW) based on estimated linkage probabilities (Hughes et al., 2019). I focus on marriage because this was the primary channel through which lottery outcome affected selection. I estimate these linkage probabilities from a logit regression predicting whether a lottery participant's subsequent marriage was found in the archive with dummy variables for each lottery outcome and initial lottery age (details in appendix). By reweighting observations according to their estimated probability of successful linkage, IPW reduces bias from differential linkage success and restores representativeness. This approach assumes data are missing at random (MAR) conditional on observed covariates and that the selection mechanism is well-defined (Little and Rubin, 2002). Because linkage success is primarily driven by name distinctiveness and record completeness—factors plausibly unrelated to fertility conditional on age and lottery outcome—these assumptions appear reasonable.

Finally, because the lottery incented earlier marriage by paying a £100 bounty, it is possible that this payment is the cause of fertility behaviour, not marriage timing. This would violate the exclusion restriction. To address this concern, I re-run the analysis in a subsample of only those who received the bounty after either one or two periods. While selection into treatment remains a possibility, this approach should minimize differences in unobservables affecting selection because the two groups differ on the smallest possible time margin.¹³

Taken together, these strategies are designed to allow credible identification of the effect of marriage age on fertility. In addition, following recent critiques that highlight how methods based on birth intervals can be prone to model misspecification, I adopt a transparent and parsimonious estimation strategy to avoid similar pitfalls (Clark and Cummins, 2019; Alter, 2019).

My preferred model is

$$(14) \quad y_{iT} = \beta_0 + \beta_1 x_i + \mathbf{A}\gamma + \varepsilon_i,$$

where y_{iT} is the number of children born to woman i before she reaches age T , x_i marriage age, and $\mathbf{A}\gamma$ is a vector of controls including initial lottery age. This approach keeps all fertility outcomes on the left-hand-side and avoids issues of serial dependency that may arise when estimating individual birth intervals. Further, because T is fixed, β_1 is naturally interpreted via the birth interval. Later marriage will lead to lower y_{iT} in the absence of a compensating change to birth spacing. A negative coefficient suggests birth spacing does not fully compensate for variation in marriage timing. In the main empirical test, I analyse $T = 30$ and $T = 35$ because the natural decline in fecundity with age is negligible before these thresholds.

The value that β_1 would take in the absence of contraceptive effort must also be identified. The demographic literature often uses Hutterite fertility to model natural fertility. This is a religious minority population that resides in segregated communities in

¹³The identifying assumptions in this case are similar to regression discontinuity.

the American Midwest and holds strong religious views forbidding contraception of any kind. When their fertility was measured in the 1940s, the median woman aged over 45 had 10.4 children. Mean inter-birth intervals were 13.55 months at age 20 and 17.92 months at age 30 (Larsen and Vaupel, 1993). This gives an annual birth rate in this natural fertility population in the range of 0.67-0.89 per annum for an age group comparable to lottery participants.

However, because underlying fecundity may differ between the Hutterites and lottery participants, I prefer to identify natural fertility from a more comparable population. Davenport's (2016) St. Martin in the Fields reconstitution dataset describes a population closest to the lottery population, but it does not reconstitute marriages. This makes it impossible with this dataset to adjust birth intervals by maternal age or to measure first birth intervals. A near alternative with full reconstitution is the Wrigley et al. (1997) dataset covering 26 non-urban parishes, a standard reference in this literature (e.g. Cinnirella, Klemp and Weisdorf, 2017).

To identify natural fertility, I assume the modal couple in this population was not practising any form of family planning. As discussed above, the primary determinant of this couple's fertility would be the duration of marriage. A secondary consideration is the influence of lactational amenorrhoea. First birth intervals are characteristically shorter than subsequent intervals because lactation following earlier births delays the return of ovulation, lengthening later intervals but not the first. The mean birth interval therefore rises with marriage duration as the weighting of the first interval declines relative to other intervals.

Note that under the maintained assumptions, $\lambda = 1/E(b)$. However, this assumes birth intervals are identically distributed and misses the difference between first and higher-order intervals. Rather than account for the resulting dependency between marriage duration and the mean birth interval, I consider instead the most conservative expected birth interval consistent with the lottery population's demographic characteristics. In this context, a birth interval resulting from assumed absence of contraceptive effort is conservative if it is longer; this implies unobserved and uncontrolled characteristics are

able to substantially reduce fecundity without requiring deliberate control. As discussed, the mean birth interval rises with marriage duration, and the longest marriages allowed by the lottery began at age 22, providing a conservative benchmark.

In the Wrigley et al. (1997) data, for couples with a bridal age of 22, the mean birth interval is 1.77 years for children born before the mother turns 30 and 2.04 years before she turns 35. These means provide testable nulls for the birth rate in the absence of contraceptive effort of 0.57 for $T = 30$ and of 0.49 for $T = 35$. If some of these couples were exerting contraceptive effort, the test would be too conservative and the true natural-fertility slope would be closer to that of the Hutterites. For lottery participants who marry later than 22, the test would also be too conservative.

Formally, these identifying assumptions motivate the test $H_0 : \beta_1 = -\lambda_{nat}$ against $H_1 : \beta_1 > -\lambda_{nat}$ where $\lambda_{nat} = 0.57$ at $T = 30$ and $\lambda_{nat} = 0.49$ at $T = 35$.

V. Results

Table 4 presents the main estimates. Specification (1) is an OLS regression with the number of children born before age 30 as the outcome. Because lottery entry age is held constant, remaining variation in marriage age comes from the delay introduced by the lottery itself. The OLS result thus identifies effect of latter marriage under the assumption that the lottery is exogenous to fertility. The negative and statistically significant coefficient on marriage age ($\beta = -0.29$, $SE = 0.05$) suggests fertility declines with later marriage, and the natural fertility hypothesis is rejected ($p = 0.00$). Specification (2) is the corresponding 2SLS estimate and instruments marriage age with first-round winning to address concerns about endogenous lottery drop-out. It is attenuated relative to the OLS estimate ($\beta = -0.18$, $SE = 0.14$), which suggests selection in the OLS regression works in favour of the natural fertility hypothesis. An intuitive explanation is that couples who dropped out of the lottery early valued fertility more than those who stayed; these couples selected into earlier marriage and went on to have larger families, making the marriage age–fertility gradient steeper than it would be had they remained in the lottery to the end. The 2SLS coefficient, which strips out this selection effect, is closer to zero

TABLE 4—MAIN RESULTS

	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
	OLS: T=30	IV: T=30	OLS: T=35	IV: T=35	OLS: T=30	IV: T=30	OLS: T=35	IV: T=35
Marriage Age	-0.29 (0.05)	-0.18 (0.14)	-0.12 (0.07)	0.019 (0.22)	-0.32 (0.05)	-0.18 (0.16)	-0.12 (0.06)	-0.043 (0.22)
Start Age	0.019 (0.06)	-0.080 (0.13)	0.0053 (0.09)	-0.11 (0.20)	0.047 (0.05)	-0.062 (0.13)	-0.0038 (0.08)	-0.063 (0.19)
Constant	8.57 (0.99)	7.98 (1.21)	5.75 (1.53)	5.04 (1.85)	8.54 (0.82)	7.69 (1.37)	5.78 (1.48)	5.32 (2.09)
H ₀ : Natural Fertility	0.00	0.01	0.00	0.02	0.00	0.02	0.00	0.05
R ²	0.41	0.38	0.048	0.015	0.43	0.38	0.050	0.040
N	103	103	103	103	102	102	102	102
First-stage F		13.52		13.52		27.49		27.49
IPW					✓	✓	✓	✓

Note: Standard errors in parentheses.

and inconsistent with the natural fertility hypothesis ($p = 0.01$).

Specification (3) increments T by five years, considering births before age 35 under OLS. The coefficient is further attenuated ($\beta = -0.12$, $SE = 0.07$) relative to fertility before age 30. This is consistent with adjustment on the effort margin being exerted evenly across marriage rather than concentrated at the beginning; the full substitution of effort for timing occurs over a long span. Specification (4) is the corresponding 2SLS estimate, which is now marginally positive but not different from zero ($\beta = 0.019$, $SE = 0.22$). Both coefficients are incompatible with the natural fertility hypothesis. The implication is qualitatively similar to columns (1) and (2). Change to marriage timing is offset by contraceptive effort, resulting in no observable difference in fertility by lottery outcome. Columns (5)-(8) mirror exactly the regressions in columns (1)-(4), now using IPW to address remaining concerns about selection related to linking rates. The conclusions do not qualitatively differ from the unweighted estimates.

In every case, the natural fertility hypothesis is rejected at conventional levels. Lottery participants' contraceptive effort adjusted to exogenous variation in marriage timing to eliminate fertility differences over time. Further, in every case, the IV coefficient is attenuated relative to its OLS pair, suggesting selection into treatment works against this conclusion. The first-stage regressions exceed conventional thresholds for weak instru-

TABLE 5—RESULTS FOR SUBSAMPLE IN RECEIPT OF MARRIAGE PORTION

	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
	OLS: T=30	IV: T=30	OLS: T=35	IV: T=35	OLS: T=30	IV: T=30	OLS: T=35	IV: T=35
Marriage Age	-0.069 (0.20)	0.49 (0.45)	0.19 (0.32)	1.48 (0.78)	-0.068 (0.19)	0.49 (0.45)	0.19 (0.30)	1.48 (0.77)
Start Age	-0.24 (0.23)	-0.87 (0.51)	-0.39 (0.37)	-1.82 (0.88)	-0.24 (0.21)	-0.87 (0.49)	-0.39 (0.33)	-1.83 (0.84)
Constant	9.34 (1.57)	10.3 (1.80)	7.87 (2.49)	10.1 (3.09)	9.34 (1.14)	10.3 (1.57)	7.87 (1.99)	10.1 (3.25)
H ₀ : Natural Fertility	0.02	0.02	0.04	0.01	0.01	0.02	0.03	0.01
R ²	0.38	0.26	0.078	.	0.38	0.26	0.078	.
N	44	44	44	44	44	44	44	44
First-stage F		11.43		11.43		11.28		11.28
IPW					✓	✓	✓	✓

Note: Standard errors in parentheses.

ments ($F > 10$) and first-stage regressions show little sign of heteroscedasticity. Residual concern about weak-instrument bias would tend to work against rejecting natural fertility since weak instruments tend to bias estimates toward OLS.

Table 5 presents estimates from a subsample who received the marriage portion after one or two lotteries. The comparison is therefore between two groups who won, married, and were paid £100 within approximately six months of each other. Unobservable differences between these two groups driven by impatience and lottery exit should therefore be minimized, and their fertility behaviour should reflect the pure effect of marginal differences in marriage timing. As before, although the signal is noisier in the smaller sample, natural fertility is rejected at conventional levels across all specifications. Specifications (5)-(8) introduce IPW to correct for sample selection driven by record linkage and do not qualitatively differ from unweighted results. In the appendix, jackknife standard errors for table 5 are presented to address concerns that individual observations might be driving the results given the small sample, and the findings appear robust.

VI. Complementary evidence: feasts

The lottery provides a novel and possibly unique natural experiment that identifies fertility control with high internal validity, yet the behaviour must be demonstrated in a

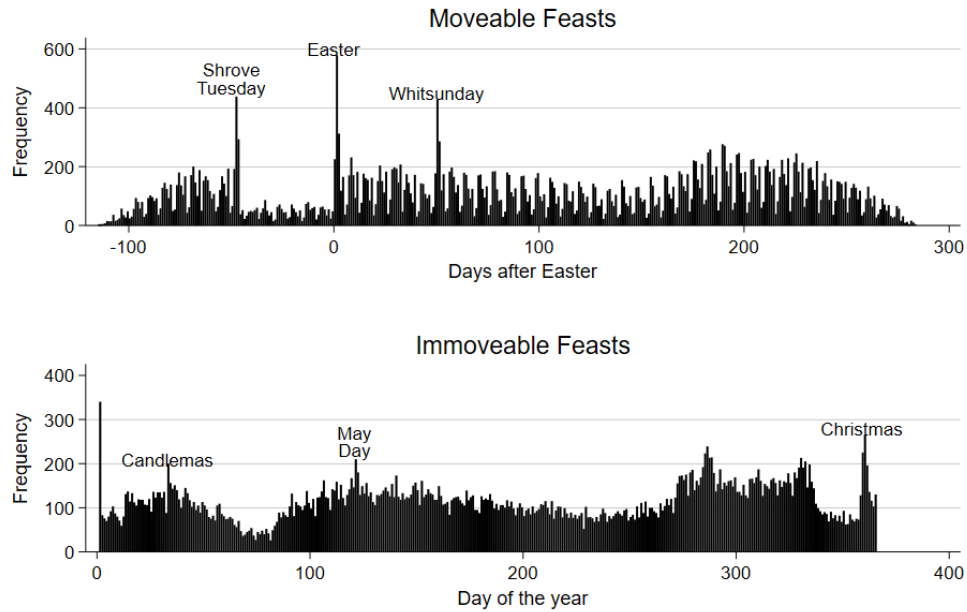


FIGURE 3. FREQUENCY OF MARRIAGE ACROSS MOVEABLE AND IMMOVABLE FEASTS (1750-1837)

Source: Wrigley et al. (1997)

Note: A very large number of marriages occurred on January 1st, but this was not obviously a feast in the past. It is more likely that this reflects transcription errors, possibly where the particular day of the marriage was illegible in the original documents and the first day of the year was chosen out of convenience.

more general population before any broad claims about family planning in pre-transition England can be sustained (Deaton and Cartwright, 2017). This section therefore presents complementary evidence, in the sense that it exchanges internal validity for greater external validity, of the substitutability of marriage timing and contraceptive effort in a larger population.

I use the family reconstitutions of Wrigley et al. (1997), which are the standard empirical data for pre-transition demography in England, analysing women whose birth histories are not truncated before age 35. This threshold provides additional power in contrast to Wrigley et al.'s method for measuring completed fertility, which takes women whose birth histories were not truncated before age 45. As with the lottery sample, this is not an analytical limitation because the test concerns the relationship between marriage timing

and contraceptive effort rather than completed lifetime fertility. I also restrict the sample to marriages occurring after 1752 to avoid the complication of the Julian-Gregorian calendar transition and better match the dates of the lottery.

For identification, I use marriage on a feast day as an instrument for marriage timing. Some couples may resolve their courtship early because they have a preference for marrying on a day of special significance within the calendar year. The identifying assumption is that feast-induced shifts in marriage timing are uncorrelated with unobserved determinants of fertility. The liturgical calendar strongly influenced the timing of marriage in England, with prohibitions against marrying during Lent and Advent (Cressy, 1985). Feast days, in contrast, were popular occasions for marriage, offering respite from work, revelry, and communal celebration (Boulton, 1989). Figure 3 shows the frequency of marriage over the moveable (paschal) calendar in the top panel and the immoveable calendar in the bottom panel. Six days stand out as particularly popular feasts for marriage: Shrove Tuesday, Easter, Whitsunday, Candlemas, May Day, and Christmas. Michaelmas (September 29; day 272) also stands out in Figure 3, but marriages at this time reflected the expiration of annual service contracts at harvest's end rather than a preference for the feast itself. Because this economic channel could directly affect fertility outcomes, I exclude Michaelmas from the instrument set. I combine the remaining six feast days into a single instrument for marriage on a popular feast.

Table 6 column (1) presents the first-stage results. Marriage on a feast day predicts brides will be roughly nine months (0.72 years) younger than those marrying at any other time during the year. I include as controls dummy variables for whether the groom was described as a labourer, which would mark him as unskilled, whether the groom was described as a gentleman, a marker of high status, and whether the first birth interval was shorter than nine months, a measure of premarital conception (PMC). In the first stage, all feasts are pooled together into the same indicator variable for greater power. In contrast, column (2) presents disaggregated results. The disaggregated coefficients are all negative and of broadly similar magnitude, with power limiting significance for individual feasts. This supports pooling.

Columns (3) and (4) consider whether marriage market outcomes differ on feast days. The outcome variable in column (3) is the spousal age difference, and the regression is unable to reject no difference in spousal age on feasts. This is consistent with couples jointly deciding to resolve their courtship earlier. Column (4) addresses a related threat from differential assortative matching. The interaction between feast and groom's age is small and insignificant ($\beta = 0.016$, $SE = 0.04$), indicating that age-assortative matching occurs similarly on feast and non-feast days.

The final balance tests are probit regressions on binary outcomes. Column (5) considers whether grooms who marry on feasts have different labour market outcomes. The Wrigley et al. (1997) dataset records the father's occupation each time it is given in the source documents. This could be at marriage, baptism of the children, or at death. The outcome variable in this column is equal to one if the father was ever described as a 'labourer' in the reconstitution file. The regression in column (5) gives no indication of difference on this measure of labour-market disadvantage. Column (6) considers whether feast-day couples differed in premarital sexuality. The outcome is a binary variable equal to one if the first birth interval of the marriage is shorter than 9 months, indicating premarital conception (PMC). One concern might be that couples who married on holidays were less likely to have married in haste to avoid an out-of-wedlock birth, but the evidence in column (6) does not support this concern. Note that the gentleman variable perfectly predicts the absence of premarital conception so is excluded from this regression.

The main test of marriage timing's effect on contraceptive effort is presented in table 7. The OLS estimate in column (1) predicts that brides who marry one year later will have 0.35 fewer births ($\beta = -0.35$, $SE = 0.00$) before age 30. Note that this result suggests spacing and marriage age are not independent and is inconsistent with the natural fertility hypothesis ($p = 0.00$). As Cinnirella, Klemp and Weisdorf (2017) have documented using this same dataset, younger (older) brides go on to have longer (shorter) birth intervals after conditioning on a rich set of covariates including parity, which would tend

TABLE 6—FIRST STAGE AND BALANCE TESTS FOR FEAST DAYS

	(1) OLS: Bride's age	(2) OLS: Bride's age	(3) OLS: Age diff.	(4) OLS: Bride's age	(5) Probit: Labourer	(6) Probit: PMC
Feast	-0.72** (0.22)		0.15 (0.35)	-0.70 (1.05)	0.12 (0.09)	0.034 (0.07)
Labourer	-0.61*** (0.17)	-0.60*** (0.17)	-0.11 (0.28)	-0.55** (0.19)		0.15** (0.05)
Gentleman	-0.035 (1.15)	-0.023 (1.15)	-3.96* (1.79)	-0.60 (1.19)		0 (.)
PMC	-0.60*** (0.13)	-0.60*** (0.13)	0.15 (0.21)	-0.29* (0.14)	0.17*** (0.05)	
Easter		-0.97* (0.40)				
Shrove Tuesday		-0.45 (0.58)				
Whitsunday		-0.56 (0.48)				
Candlemas		-0.64 (0.99)				
May Day		-0.53 (0.91)				
Christmas		-0.76 (0.44)				
Groom's age				0.26*** (0.01)		
Feast × Groom's age				0.016 (0.04)		
R2	0.054	0.054	0.016	0.18		
N	4698	4698	3526	3526	4184	4684

Note: Standard errors in parentheses. PMC stands for pre-marital conception. Age difference is measured as the bride's age minus the groom's. Fixed effects for month of the year, parish, and a quadratic time trend are estimated but not shown.

to flatten the gradient between marriage age and fertility.¹⁴ This is entirely consistent with substitution between marriage timing and contraceptive effort, but it may also be attributable to unobservable characteristics related to bridal age.

Column (2) presents the corresponding IV estimates using feast-days. Assuming pref-

¹⁴Maternal age at marriage is included as an unreported control variable in their paper, but this result is available in the replication package (Cinnirella, Klemp and Weisdorf, 2017).

TABLE 7—MARRIAGE TIMING AND CONTRACEPTIVE EFFORT IN RECONSTITUTION DATA

	(1)	(2)	(3)	(4)
	OLS:	IV:	OLS:	IV:
	T=30	T=30	T=35	T=35
Bride's age	-0.35 (0.00)	-0.21 (0.08)	-0.40 (0.00)	-0.28 (0.10)
Labourer	-0.070 (0.04)	0.017 (0.07)	-0.17 (0.05)	-0.10 (0.08)
Gentleman	0.12 (0.24)	0.12 (0.29)	0.24 (0.33)	0.24 (0.36)
PMC	0.50 (0.03)	0.58 (0.06)	0.57 (0.04)	0.63 (0.07)
H_0 : Natural Fertility	0.00	0.00	0.00	0.03
R ²	0.76	0.65	0.68	0.63
N	4698	4698	4698	4698
First-stage F		10.26		10.26

Note: Standard errors in parentheses. Fixed effects for month of the year, parish, and a quadratic time trend are estimated but not shown.

erence for feast-day weddings is orthogonal to fertility, the coefficient ($\beta = -0.21$, $SE = 0.08$) identifies how earlier marriage affects fertility directly. Although feast-day brides who marry early will have more children, the predicted difference is attenuated relative to OLS and inconsistent with the hypothesis that they do not exert contraceptive effort ($p = 0.00$). Feast-day couples appear to offset the higher fertility that would otherwise result from their earlier marriage. Columns (3) and (4) parallel exactly columns (1) and (2), except they consider fertility before age 35 as the outcome. The results are qualitatively unchanged. Natural fertility is rejected across all specifications.

Although the identifying assumptions in this analysis are weaker, these results should be read alongside the lottery analysis. Both analyses demonstrate that the effect of marriage age on fertility is weaker than expected under the assumption that couples did not exert contraceptive effort. In the lottery analysis, this finding is cleanly identified in a small population whose behaviour may not generalise; in the feast-day analysis, it is identified less cleanly in a standard reference population. The feast-day evidence also identifies this result without a concurrent wealth shock, which works with the winners-only analysis in Table 5 to rule out wealth as a violation of the exclusion restriction in the lottery. Together, both analyses converge on the same conclusion: contraceptive effort

and marriage timing were substitute family planning strategies.

VII. Conclusion

This paper demonstrates a causal connection between two mechanisms that have so far been only suggestively connected in the literature (Szreter and Garrett, 2000; Guinnane, 2011; Voigtländer and Voth, 2013). While variation in marriage timing explains a greater share of variation in crude fertility rates before the fertility transition, researchers have tended to rely on short-term changes in birth spacing following income shocks to identify family planning (e.g. van Bavel, 2004a; Cinnirella, Klemp and Weisdorf, 2017). The evidence above shows that an exogenous change to the dominant mechanism brings about a compensating change in the second; they were substitutes.

Where two alternatives are substitutes, the less costly option will tend to be preferred. As delayed marriage was a practice dating back to the medieval period in England, it was unlikely to be subject to social sanctions, and labour markets for single women were well-established (Horrell, Humphries and Weisdorf, 2020; Humphries and Weisdorf, 2015; Voigtländer and Voth, 2013). Plausibly, it was less costly to delay marriage than to exert high contraceptive effort within marriage. However, as the model demonstrates, substitutability can break down when the marginal value of children declines and the expected number of children falls.

In theory, this dynamic may correspond with a rise in the value of child quality relative to quantity resulting from skill-biased technological change (Galor and Weil, 2000). A recent paper demonstrates that the extent of child labour in factories in 1851 has been underestimated by a third (You and Tertzakian, 2026), suggesting technological change had not yet eliminated the advantage of child quantity on the eve of the fertility transition. A number of developments may have finally pushed the balance in favour of quality at the end of the nineteenth century: technological change associated with the second industrial revolution (O'Rourke, Rahman and Taylor, 2013), a series of reforms reducing the costs of schooling (Milner, 2025), or public health investment raising the lifetime return on human capital (Aidt, Davenport and Gray, 2023). Once this occurred, the model

predicts that substitutability between contraceptive strategies may become complementarity, and both delayed marriage and greater effort may be activated to achieve lower fertility. There is evidence that this occurred in England at the end of the nineteenth century (Woods, 2000). Changes in the techniques used to reduce fertility are therefore anticipated by the model and do not require agents to learn new behaviours. Indeed, the model anticipates that innovations improving the efficiency of contraceptive effort may re-activate substitutability, which is also consistent with early twentieth-century evidence in England (Woods, 2000; Jones, 2020). On a broader level, therefore, this paper establishes a long continuity between pre-transition and post-transition contraception (Bailey, 2025).

REFERENCES

- Aidt, Toke S., Romola J. Davenport, and Felix Gray.** 2023. “New perspectives on the contribution of sanitary investments to mortality decline in English cities, 1845-1909.” *Economic History Review*, 76: pp. 624–60.
- Alter, George C.** 2019. “The evolution of models in historical demography.” *Journal of Interdisciplinary History*, 50(3): pp. 325–62.
- Anderson, Siwan, and Chris Bidner.** 2023. “An institutional perspective on the economics of the family.” In *Handbook of the Economics of the Family*. Vol. 1, , ed. Shelly Lundberg and Alessandra Voena, 443–500. Elsevier.
- Arroyo, Cristino R., and Junsen Zhang.** 1997. “Dynamic microeconomic models of fertility choice: a survey.” *Journal of Population Economics*, 10: pp. 23–65.
- Bailey, Martha J.** 2025. “Economics of childbearing: trends, progress, and challenges.” Working Paper 33970, Cambridge, Mass.
- Bailey, Martha J., Connor Cole, Morgan Henderson, and Catherine Massey.** 2020. “How well do automated linking methods perform? lessons from US historical data.” *Journal of Economic Literature*, 58(4): pp. 997–1044.

- Beach, Brian, and W. Walker Hanlon.** 2023. "Culture and the historical fertility transition." *Review of Economics Studies*, 90(4): pp. 1669–1700.
- Bengtsson, Tommy, and Luciana Quaranta.** 2025. "The escape from hunger: the impact of food prices on well-being in Sweden, 1813-1967." *Explorations in Economic History*, 96: 101652.
- Bengtsson, Tommy, and Martin Dribe.** 2006. "Deliberate control in a natural fertility population: Southern Sweden, 1766-1864." *Demography*, 43(4): pp. 727–46.
- Blanc, Guillaume.** 2024. "The cultural origins of the demographic transition in France." unpublished working paper.
- Blanc, Guillaume, and Romain Wacziarg.** 2020. "Change and persistence in the age of modernization: Saint-Germain-d'Anxure, 1730-1895." *Explorations in Economic History*, 78: 101352.
- Bongaarts, John.** 2020. "Trends in fertility and fertility preferences in sub-Saharan Africa: the roles of education and family planning programs." *Genus*, 76(32).
- Boot, H. M.** 1995. "How Skilled Were Lancashire Cotton Factory Workers in 1833." *Economic History Review*, 48(2): 283–303.
- Boulton, Jeremy.** 1989. "Economy of time? Wedding days and the working week in the past." *Local Population Studies*, 43: pp. 28–46.
- Braun, Sebastian T., Richard Franke, and Timur Öztürk.** 2025. "Protestantism, Industrialization, and the Fertility Decline: Evidence from Religious Enclaves in Württemberg, 1871–1930."
- Burnette, Joyce.** 2006. "How skilled were English agricultural labourers in the early nineteenth century?" *Economic History Review*, 59(4): pp. 688–716.
- Carlsson, Gösta.** 1966. "The decline of fertility: innovation or adjustment process." *Population Studies*, 20(2): pp. 149–74.

- Cavalcanti, Tiago, Georgi Kocharkov, and Cezar Santos.** 2021. “Family planning and development: aggregate effects of contraceptive use.” *Family planning and development: aggregate effects of contraceptive use*, 131(634): 624–57.
- Cervellati, Matteo, Gerrit Meyerheim, and Uwe Sunde.** 2023. “The empirics of economic growth over time and across nations: a unified growth perspective.” *Journal of Economic Growth*, 28: pp. 173–224.
- Cilliers, Jeanne, Martine Mariotti, and Igor Martins.** 2024. “Fertility responses to short-term economic stress: price volatility and wealth shocks in a pre-transitional settler colony.” *Explorations in Economic History*, 94: 101620.
- Cinnirella, Francesco, Marc Klemp, and Jacob Weisdorf.** 2017. “Malthus in the bedroom: birth spacing as birth control in pre-transition England.” *Demography*, 54(2).
- Cinnirella, Francesco, Marc Klemp, and Jacob Weisdorf.** 2019. “Further evidence of within-marriage fertility control in pre-transitional England.” *Demography*, 56(4): pp. 1557–72.
- Clark, Gregory.** 2007. *A farewell to alms: a brief economic history of the world*. Princeton: Princeton University Press.
- Clark, Gregory, and Neil Cummins.** 2015. “Malthus to modernity: wealth, status, and fertility in England, 1500-1879.” *Journal of Population Economics*, 28: pp. 3–29.
- Clark, Gregory, and Neil Cummins.** 2019. “Randomness in the bedroom: there is no evidence for fertility control in pre-industrial England.” *Demography*, 56(4): pp. 1541–55.
- Clark, Gregory, Neil Cummins, and Matthew Curtis.** 2020. “Twins support the absence of parity-dependent fertility control in pretransition populations.” *Demography*, 57(4): pp. 1571–95.
- Coale, Ansley J.** 1973. “The demographic transition reconsidered.” Vol. 1, pp. 53–72. Liège, Belgium.

- Coale, Ansley J., and Roy Treadway.** 1986. "A Summary of the Changing Distribution of Overall Fertility, Marital Fertility, and the Proportion Married in the Provinces of Europe." In *The decline of fertility in Europe.*, ed. Ansley J. Coale, pp. 31–181. Princeton:Princeton University Press.
- Cockburn, J. S., H. P. F. King, and K. G. T. McDonnell,** ed. 1969. *A history of the county of Middlesex.* London:Victoria County History.
- Colejo-Durán, Lidia, Fanie Pelletier, Lisa Dillon, Alain Gagnon, and Patrick Bergeron.** 2024. "Early and adult life environmental effects on reproductive performance in preindustrial women." *PLoS ONE*, 19(10): e0290212.
- Conde-Aguedelo, Agustín, Anyeli Rosas-Bermudez, Fabio Castaño, and Maureen H. Norton.** 2012. "Effects of birth spacing on maternal, perinatal, infant, and child health: a systematic review of causal mechanisms." *Studies in Family Planning*, 43(2): 93–114.
- Cook, Hera.** 2004. *The long sexual revolution: English women, sex, and contraception 1800-1975.* Oxford:Oxford University Press.
- Cressy, David.** 1985. "The seasonality of marriage in old and new England." *Journal of Interdisciplinary History*, 16(1): pp. 1–21.
- Dadi, Abel Fekadu.** 2015. "A systematic review and meta-analysis of the effect of short birth interval on infant mortality in Ethiopia." *PLoS ONE*, 10(5): e0126759.
- Davenport, Romola.** 2016. "Urban family reconstitution—a worked example." *Local Population Studies*, 96: pp. 28–49.
- Davenport, Romola J.** 2019. "Infant-feeding practices and infant survival by familial wealth in London, 1752-1812." *History of the Family*, 24(1): pp. 174–206.
- Deaton, Angus, and Nancy Cartwright.** 2017. "Understanding and misunderstanding randomized controlled trials." *Social Science & Medicine*, 210: pp. 2–21.

- de la Croix, David, and Faustine Perrin.** 2018. “How far can economic incentives explain the French fertility and education.” *European Economic Review*, 108: pp. 221–45.
- de la Croix, David, Eric B. Schneider, and Jacob Weisdorf.** 2019. “Childlessness, celibacy and net fertility in pre-industrial England: the middle-class evolutionary advantage.” *Journal of Economic Growth*, 24(3): pp. 223–56.
- de Silva, Tiloka, and Silvana Tenreyro.** 2020. “The fall in global fertility.” *American Economic Journal: Macroeconomics*, 12(3): pp. 77–109.
- Dillon, Lisa, Alla Chernenko, Martin Dribe, Sacha Engelhardt, Alain Gagnon, Heidi A. Hanson, Huong Meeks, Luciana Quaranta, Ken R. Smith, and Hélène Vézina.** 2024. “Did grandmothers enhance reproductive success in historic populations?: testing evolutionary theories on historical demographic data in Scandinavia and North America.” In *Human Evolutionary Demography*. , ed. Oskar Burger, Ronald Lee and Rebecca Sear, pp. 476–502. Open Book Publishers.
- Dioikitopoulos, Evangelos, and Dimitros Varvarigos.** 2023. “Delay in childbearing and the evolution of fertility rates.” *Journal of Population Economics*, 36: pp. 1545–71.
- Doepke, Matthias, Anne Hannusch, Fabian Kindermann, and Michèle Tertilt.** 2023. “The economics of fertility: a new era.” In *Handbook of the economics of the family*. Vol. 1, pp. 151–254. Elsevier.
- Folbre, Nancy.** 2009. *Greed, lust & gender: a history of economic ideas*. Oxford:Oxford University Press.
- Foreman-Peck, James.** 2011. “The Western European marriage pattern and economic development.” *Explorations in Economic History*, 48: 292–309.
- Galor, Oded, and David N. Weil.** 2000. “Population, Technology, and Growth: From Malthusian Stagnation to the Demographic Transition and Beyond.” *American Economic Review*, 90(4): pp. 806–828.

- Galor, Oded, and Omer Moav.** 2002. "Natural selection and the origin of economic growth." *Quarterly Journal of Economics*, 117(4): 1133–1191.
- Gay, Victor, Paula Gobbi, and Marc Goñi.** 2026. "Revolutionary transition: inheritance change and fertility decline." *Journal of Political Economy*, forthcoming.
- Goldin, Claudia.** 2021. *Career and family: women's century-long journey toward equity*. Princeton:Princeton University Press.
- Greenwood, Jeremy, Nezih Guner, and Guillaume Vandenbroucke.** 2017. "Family Economics Writ Large." *Journal of Economic Literature*, 55(4): 1346–1434.
- Guinnane, Timothy W.** 2011. "The Historical Fertility Transition: A Guide for Economists." *Journal of Economic Literature*, 49(3): 589–614.
- Guo, Guang, and Laurence M. Grummer-Strawn.** 1993. "Child mortality among twins in less developed countries." *Population Studies*, 47(3): pp. 495–510.
- Henry, Louis.** 1961. "Some data on natural fertility." *Eugenics Quarterly*, 8(2): pp. 81–91.
- Horrell, Sara, Jane Humphries, and Jacob Weisdorf.** 2020. "Malthus's missing women and children: demography and wages in historical perspective, England 1280-1850." *European Economic Review*, 129: 103534.
- Horrell, Sara, Jane Humphries, and Jacob Weisdorf.** 2022. "Beyond the male breadwinner: life-cycle living standards of intact and disrupted English working families, 1260-1850." *Economic History Review*, 75(2): pp. 530–60.
- Hruschka, Daniel J., and Oskar Burger.** 2016. "How does variance in fertility change over the demographic transition?" *Philosophical transactions of the Royal Society of London. Series B, Biological sciences*, 371: 20150155.
- Hughes, Rachel A., Neil M. Davies, George Davey Smith, and Kate Tilling.** 2019. "Selection bias when estimating average treatment effects using one-sample instrumental variable analysis." *Epidemiology*, 30(3): pp. 350–7.

- Humphries, Jane, and Jacob Weisdorf.** 2015. "The Wages of Women in England, 1260-1850." *Journal of Economic History*, 75(2): 405–447.
- Jones, Claire L.** 2020. *The business of birth control: contraception and commerce in Britain before the sexual revolution*. Manchester:Manchester University Press.
- Jones, Harry.** 1875. *East and west London: being notes of common life and pastoral work in Saint James's, Westminster and in Saint Georges'-in-the-East*. London:Smith, Elder & Co.
- Jones, M. G.** 1964. *The Charity School Movement: A Study of Eighteenth Century Puritanism in Action*. London:Frank Cass and Co.
- Kaiser, Moritz.** 2025. "Wage determination and employer power in the labour market for servants: evidence from England and Wales, 1780-1834." *International Review of Social History*, 70(S33): pp. 81–105.
- Keeley, Michael C.** 1977. "The economics of family formation." *Economic Inquiry*, 15(2): pp. 238–50.
- Klesment, Martin, and Kersti Lust.** 2025. "The fertility response to price changes in a manorial society: the case of rural Estonia, 1834-1884." *Explorations in Economic History*, 96: 101653.
- Larsen, Ulla, and James W. Vaupel.** 1993. "Hutterite fecundability by age and parity: strategies for frailty modeling of event histories." *Demography*, 30(1): pp. 81–102.
- Le Fur, Tanguy, and Etienne Wasmer.** 2025. "Fighting for resources: a unified growth model of the Great Divergence." *Journal of Economic Growth*, Online first: n.p.
- Lesthaeghe, Ron J.** 1977. *The decline of Belgian fertility, 1800-1970*. Princeton:Princeton University Press.
- Lincoln, Margarette.** 2018. *Trading in War: London's Maritime World in the Age of Cook and Nelson*. New Haven:Yale UP.

- Little, Roderick, and Donald Rubin.** 2002. *Statistical analysis with missing data*. Hoboken, N.J.:Wiley-Interscience.
- Lundberg, Shelly, and Robert A. Pollak.** 2007. “The American family and family economics.” *Journal of Economic Perspectives*, 21(2): 3–26.
- Madsen, Jakob B., Md. Rabiul Islam, and Xueli Tang.** 2020. “Was the post-1870 fertility transition a key contributor to growth in the West in the twentieth century?” *Journal of Economic Growth*, 25: 431–54.
- Malthus, Thomas. R.** 1909. *Parallel chapters from the first and second editions of an essay on the principle of population*. New York:Macmillan Company.
- Marco-Gracia, Francisco J.** 2019. “Fertility control due to short-term economic stress in rural Aragón (Spain), 1801-1909.” *Journal of Iberian and Latin American Economic History*, 39(1): 129–55.
- Marco-Gracia, Francisco J.** 2024. “Trends in twin births and survival in rural Spain: evidence from 18th to 20th century.” *Twin Research and Human Genetics*, 27(6): 302–10.
- Marriott, John.** 2011. *Beyond the Tower: A History of East London*. London:Yale UP.
- McFadden, Clare.** 2023. “From the ground up: a multidisciplinary approach to past fertility and population narratives.” *Human Nature*, 34: pp. 476–500.
- Measuringworth.com.** 2024. “Relative values - UK £.”
- Melki, Mickael, Hillel Rapoport, Enrico Spolaore, and Romain Wacziarg.** 2024. “Cultural remittances and modern fertility.” Working Paper 32990, Cambridge, Mass.
- Miles, Dudley.** 1988. *Francis Place, 1771-1854: the life of a remarkable radical*. Sussex:Harvester Press.
- Miller, Grant, and Kimberly Singer Babiarz.** 2016. “Family planning program effects: evidence from microdata.” *Population and Development Review*, 42(1): 7–26.

- Milner, Benjamin.** 2025. “The impact of state-provided education: evidence from the 1870 education act.” *Economic Journal*, 135(671): pp. 2302–2337.
- Moorthy, Vivek, Lakshmi Iyer, and Paloma Moyano.** 2025. “Religion and demography: papal influences on fertility.”
- Munshi, Kaivan, and Myaux.** 2006. “Social norms and the fertility transition.” *Journal of Development Economics*, 80: pp. 1–38.
- North, Douglass C.** 1990. *Institutions, institutional change and economic performance*. Cambridge:Cambridge University Press.
- Oris, Michel, Stanislao Mazzoni, and Diego Ramiro-Fariñas.** 2024. “Did the 1917-21 economic depression accelerate the epidemiological transition? milk prices, summer peak of mortality, and food-and-water causes of death in Madrid, Spain.” *Explorations in Economic History*, 94: 101613.
- O’Rourke, K., A. Rahman, and M. Taylor.** 2013. “Luddites, the Industrial Revolution, and the Demographic Transition.” *Journal of Economic Growth*, 18(4): 373–409.
- Perrin, Faustine.** 2022. “On the origins of the demographic transition: rethinking the European marriage pattern.” *Cliometrica*, 16(3): pp. 431–75.
- Pollak, Robert A.** 1985. “A transaction cost approach to families and households.” *Journal of Economic Literature*, 23(2): 581–608.
- Raine, Henry.** 1748. *Copy of an indenture of release of several freehold and leasehold premises from the late Mr. Henry Raine*. London:J. Owen. Gale Primary Sources.
- Rose, Craig.** 1991. “Evangelical philanthropy and Anglican revival: the charity schools of Augustan London, 1698-1740.” *The London Journal*, 16(1): pp. 35–65.
- Sandström, Glenn, and Lotta Vikström.** 2015. “Sex preference for children in German villages during the fertility transition.” *Population Studies*, 69(1): 57–71.

- Sarti, Raffaella.** 2008. “‘All masters discourage the marrying of their male servants, and admit not by any means the marriage of the female’: Domestic service and celibacy in Western Europe from the Sixteenth to the Nineteenth Century.” *European History Quarterly*, 38(3): 417–49.
- Scott, Susan, and C. J. Duncan.** 1999. “Nutrition, fertility and steady-state population dynamics in a pre-industrial community in Penrith, Northern England.” *Journal of Biosocial Science*, 31(4): pp. 505–23.
- Sear, Rebecca.** 2015. “Evolutionary contributions to the study of human fertility.” *Population Studies*, 69(sup1): s39–s55.
- Snell, K. D. M., and Paul S. Ell.** 2004. *Rival Jerusalems: The Geography of Victorian Religion*. Cambridge:Cambridge UP.
- Spolaore, Enrico, and Romain Wacziarg.** 2022. “Fertility and modernity.” *Economic Journal*, 132(642): pp. 796–833.
- Stanford, Joseph B., and David B. Dunson.** 2007. “Effects of sexual intercourse patterns in time to pregnancy studies.” *American Journal of Epidemiology*, 165(9): pp. 1088–95.
- Szreter, Simon.** 1996. *Fertility, class and gender in Britain, 1860-1940*. Cambridge:Cambridge University Press.
- Szreter, Simon, and Eilidh Garrett.** 2000. “Reproduction, compositional demography, and economic growth: family planning in England long before the fertility decline.” *Population and Development Review*, 26(1): pp. 45–80.
- Szreter, Simon, and Kevin Siena.** 2021. “The pox in Boswell’s London: an estimate of the extent of syphilis infection in the metropolis in the 1770s.” *Economic History Review*, 74(2): pp. 372–99.
- The London Archive.** 1736. “Raine’s foundation school.” ACC/1811.

- The London Archive.** 2010*a*. “London, England, Church of England Baptisms, Marriages and Burials, 1538-1812.”
- The London Archive.** 2010*b*. “London, England, Church of England Births and Baptisms, 1813-1923.”
- Tong, Stephen, and Roger V. Short.** 1998. “Dizygotic twinning as a measure of human fertility.” *Human Reproduction*, 13(1): pp. 95–8.
- van Bavel, Jan.** 2004*a*. “Deliberate birth spacing before the fertility transition in Europe: evidence from nineteenth-century Belgium.” *Population Studies*, 58(1): pp. 95–107.
- Van Bavel, Jan.** 2004*b*. “Diffusion effects in the European fertility transition: historical evidence from within a Belgian town (1846-1910).” *European Journal of Population*, 20(1): pp. 63–85.
- Voigtländer, Nico, and Hans-Joachim Voth.** 2013. “How the west ”invented” fertility restriction.” *American Economic Review*, 103(6): pp. 2227–64.
- Willführ, Kai P., and Josep Sottile Perez.** 2025. “The aftermath of the February flood of 1825: social and demographic change in the Krummhörn Region, East Frisia.” *Explorations in Economic History*, 96: 101650.
- Woods, Robert.** 2000. *The demography of Victorian England and Wales*. Cambridge:Cambridge University Press.
- Wrigley, E. A., and R. S. Schofield.** 1983. “English population history from family reconstitution: summary results 1600-1799.” *Population Studies*, 37: pp. 157–84.
- Wrigley, E. A., R. S. Davies, J. E. Oeppen, and R. S. Schofield.** 1997. *English Population History from Family Reconstitution 1580-1837*. Cambridge:CUP.
- Youssef, H.** 1993. “The history of the condom.” *Journal of the Royal Society of Medicine*, 86: pp. 226–8.

You, Xuesheng. 2024. “Female relatives and domestic service in nineteenth-century England and Wales: female kin servants revisited.” *Economic History Review*, 77(2): pp. 444–71.

You, Xuesheng, and Alexander Tertzakian. 2026. “Child labour and industrialization: evidence from factory records and the 1851 British census.” *Economic History Review*, 79: pp. 163–88.

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Őri, Péter, and Levente Pakot. 2025. “Fertility and mortality responses to short-term economic stress: evidence from two Hungarian sample populations, 1819-1914.” *Explorations in Economic History*, 97: 101671.

APPENDIX

A1. Additional details

INVERSE PROBABILITY WEIGHTS

Lottery outcome affects the probability of selection because the grooms of women who receive the marriage portion are known with certainty, while marriage records of those who leave early or turn down the portion must be found using only the bride's name. This leads to a higher probability of successfully linking recipients of the portion, as shown in table A1. Inverse probability weights (IPW) give observations that were unlikely to have been observed more weight in the regression to restore the representativeness of the sample. In practice, IPWs are often normalised to mitigate variance inflation due to large weights. I adopt this approach, weighting each observation by

$$\hat{w}_i = \frac{w_i}{\sum_{j=1}^n w_j} n$$

where $w_i = 1/p_i$, p_i is the estimated probability of selection, and n is sample size.

However, I do not observe the fertility of couples who have no children, either because such couples had no children or because I was unable to locate their baptismal records. If missing children were due only to lost baptism records, missingness might well be random conditional on having successfully linked the parents' marriage records. In practice, however, I am less likely to find the children of lottery candidates who won but did not immediately marry ('unmarried winners') even after they eventually married (table A1).

As de la Croix, Schneider and Weisdorf (2019) document, lifelong celibacy and marital childlessness was relatively common in England circa 1600-1840. I assume such women are overrepresented among unmarried winners, who participated in the lottery without the immediate intention to marry, and that lottery outcome did not affect whether these women eventually had children.¹⁵

¹⁵This assumption aligns with the standard no-defiers condition in instrumental variables. I treat these women as 'never-takers'.

TABLE A1—IPW SELECTION EQUATIONS

	(1) Logit: Marriage only	(2) Logit: Marriage and children
Married Winner	3.06 (0.54)	-0.074 (0.34)
Unmarried Winner	-1.54 (0.52)	-2.44 (0.71)
Start Age	0.0082 (0.03)	-0.018 (0.02)
Constant	0.089 (0.81)	0.57 (0.62)
Pseudo-R2	0.40	0.071
N	227	222

Note: Standard errors in parentheses. Dropped out of lottery before winning is omitted category.

This implies that selection based on child missingness is exogenous to the instrument and the IPWs used in the main analysis remain appropriate. Nonetheless, I re-estimate the results using a broader definition of selection that includes both marriage and child missingness (appendix tables A6 and A7). The results are qualitatively unchanged.

ADDITIONAL FIGURES AND TABLES

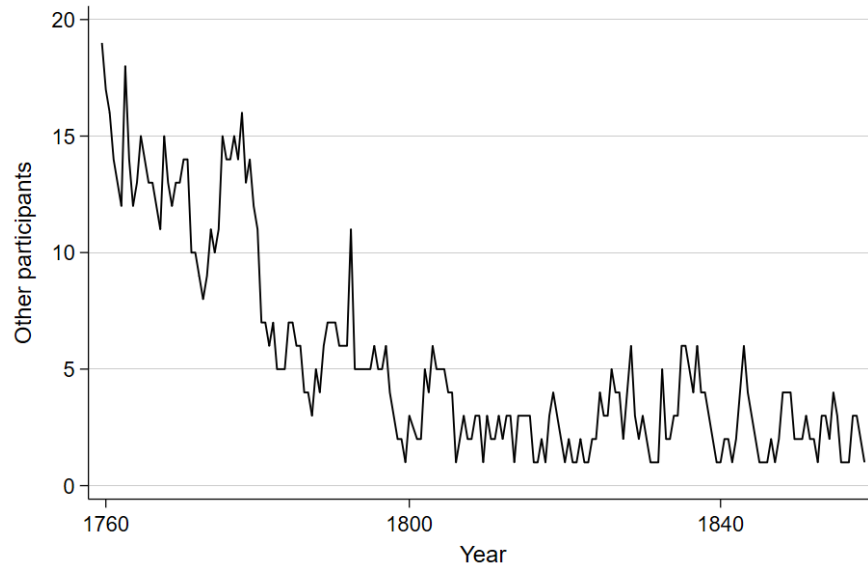


FIGURE A1. NUMBER OF PARTICIPANTS IN RAINE'S MARRIAGE PORTION CEREMONY, 1758-1872

The figure shows the number of participants by year.

TABLE A2—BIRTH INTERVALS BY WIFE’S AGE AT MARRIAGE AND DURATION OF MARRIAGE

Wife’s age at marriage	Duration of marriage (years)				
	0-4	5-9	10-4	15-9	20-4
20-4	22.02	32.38	32.95	34.29	36.76
25-9	22.65	33.34	35.98	36.27	21.03
30+	22.83	33.84	38.54	33.50	—

Source: Wrigley et al. (1997)

Prior information on birth intervals from Wrigley et al. (1997)

A2. *Robustness*

TABLE A3—MAIN ANALYSIS WITHOUT IMPUTING MISSING BIRTHS

	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
	OLS: T=30	IV: T=30	OLS: T=35	IV: T=35	OLS: T=30	IV: T=30	OLS: T=35	IV: T=35
Marriage Age	-0.25 (0.06)	-0.17 (0.16)	-0.090 (0.08)	-0.047 (0.23)	-0.29 (0.05)	-0.18 (0.18)	-0.085 (0.06)	-0.10 (0.23)
Start Age	0.025 (0.07)	-0.044 (0.15)	0.032 (0.10)	-0.0054 (0.21)	0.062 (0.06)	-0.024 (0.15)	0.024 (0.08)	0.038 (0.19)
Constant	7.31 (1.15)	6.90 (1.39)	4.16 (1.69)	3.93 (2.02)	7.29 (1.05)	6.59 (1.57)	4.14 (1.62)	4.26 (2.24)
R2	0.27	0.26	0.016	0.013	0.30	0.27	0.017	0.016
N	103	103	103	103	103	103	103	103
First-stage F		13.52		13.52		28.88		28.88
IPW					✓	✓	✓	✓

Note: Standard errors in parentheses.

This regression excludes missing births imputed from long intervals and repeats the main analysis (table 4). Estimated coefficients tend to be closer to zero, most likely due to attenuation bias from missing births, and are qualitatively comparable to the main results.

TABLE A4—SUB-SAMPLE ANALYSIS WITHOUT IMPUTING MISSING BIRTHS

	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
	OLS: T=30	IV: T=30	OLS: T=35	IV: T=35	OLS: T=30	IV: T=30	OLS: T=35	IV: T=35
Marriage Age	-0.060 (0.23)	0.47 (0.51)	0.20 (0.35)	1.32 (0.80)	-0.060 (0.20)	0.47 (0.51)	0.20 (0.31)	1.32 (0.79)
Start Age	-0.21 (0.27)	-0.80 (0.58)	-0.36 (0.40)	-1.61 (0.90)	-0.21 (0.22)	-0.80 (0.56)	-0.36 (0.33)	-1.61 (0.87)
Constant	8.37 (1.83)	9.31 (2.04)	6.86 (2.70)	8.83 (3.17)	8.37 (1.33)	9.31 (1.75)	6.86 (2.16)	8.83 (3.26)
R2	0.26	0.17	0.047	.	0.26	0.17	0.047	.
N	44	44	44	44	44	44	44	44
First-stage F		11.43		11.43		11.28		11.28
IPW					✓	✓	✓	✓

Note: Standard errors in parentheses.

This regression excludes missing births imputed from long intervals and repeats the sub-sample analysis (table 5). The sample therefore consists of only those who won on their first attempt and those who won on their second attempt. Estimated coefficients tend to be closer to zero, most likely due to attenuation bias from missing births, and are qualitatively comparable to the results in the main paper.

TABLE A5—SUB-SAMPLE RESULTS WITH JACKKNIFE STANDARD ERRORS

	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
	OLS: T=30	IV: T=30	OLS: T=35	IV: T=35	OLS: T=30	IV: T=30	OLS: T=35	IV: T=35
Marriage Age	-0.069 (0.25)	0.49 (0.51)	0.19 (0.39)	1.48 (0.89)	-0.069 (0.25)	0.49 (0.50)	0.19 (0.39)	1.48 (0.89)
Start Age	-0.24 (0.26)	-0.87 (0.54)	-0.39 (0.40)	-1.82 (0.95)	-0.24 (0.26)	-0.87 (0.54)	-0.39 (0.40)	-1.83 (0.94)
Constant	9.34 (1.17)	10.3 (1.79)	7.87 (2.07)	10.1 (3.87)	9.34 (1.17)	10.3 (1.79)	7.87 (2.07)	10.1 (3.85)
R ²	0.38	0.26	0.078	.	0.38	0.26	0.078	.
N	44	44	44	44	44	44	44	44
First-stage F		10.3		10.3		10.3		10.3
IPW					✓	✓	✓	✓

Note: Jackknife standard errors in parentheses.

This regression repeats the sub-sample analysis with jackknife standard errors due to small sample size. The sample therefore consists of only those who won on their first attempt and those who won on their second attempt. Natural fertility ($\beta = -0.40$) is rejected with at least 90-percent confidence in all IV regressions.

TABLE A6—MAIN ANALYSIS WITH ALTERNATIVE IPW WEIGHTS

	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
	OLS: T=30	IV: T=30	OLS: T=35	IV: T=35	OLS: T=30	IV: T=30	OLS: T=35	IV: T=35
Marriage Age	-0.29 (0.05)	-0.18 (0.14)	-0.12 (0.07)	0.019 (0.22)	-0.36 (0.05)	-0.15 (0.18)	-0.15 (0.06)	-0.031 (0.23)
Start Age	0.019 (0.06)	-0.080 (0.13)	0.0053 (0.09)	-0.11 (0.20)	0.099 (0.07)	-0.063 (0.13)	0.049 (0.08)	-0.040 (0.17)
Constant	8.57 (0.99)	7.98 (1.21)	5.75 (1.53)	5.04 (1.85)	8.46 (0.87)	6.75 (2.02)	5.43 (1.53)	4.49 (2.56)
R2	0.41	0.38	0.048	0.015	0.46	0.33	0.060	0.029
N	103	103	103	103	102	102	102	102
First-stage F		13.52		13.52		15.78		15.78
IPW					✓	✓	✓	✓

Note: Standard errors in parentheses.

This regression repeats the main analysis using alternative IPW weights.

TABLE A7—SUB-SAMPLE ANALYSIS WITH ALTERNATIVE IPW WEIGHTS

	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
	OLS: T=30	IV: T=30	OLS: T=35	IV: T=35	OLS: T=30	IV: T=30	OLS: T=35	IV: T=35
Marriage Age	-0.069 (0.20)	0.49 (0.45)	0.19 (0.32)	1.48 (0.78)	-0.077 (0.19)	0.47 (0.45)	0.17 (0.30)	1.44 (0.77)
Start Age	-0.24 (0.23)	-0.87 (0.51)	-0.39 (0.37)	-1.82 (0.88)	-0.23 (0.20)	-0.85 (0.48)	-0.37 (0.32)	-1.78 (0.83)
Constant	9.34 (1.57)	10.3 (1.80)	7.87 (2.49)	10.1 (3.09)	9.35 (1.13)	10.3 (1.54)	7.83 (1.96)	10.0 (3.18)
R2	0.38	0.26	0.078	.	0.39	0.27	0.079	.
N	44	44	44	44	44	44	44	44
First-stage F		11.43		11.43		11.06		11.06
IPW					✓	✓	✓	✓

Note: Standard errors in parentheses. Not qualitatively different.

This regression repeats the sub-sample analysis using alternative IPW weights.